

AI API Cost Comparison Report

SmartLabs Essay Scoring System — Monthly Cost Analysis

Generated: May 2026 | 40 Students | 40 Uses/Student/Month

1. Usage Assumptions

Total Students	40
Uses Per Student / Month	40
Total API Calls / Month	1,600
Input Tokens Per Call	~7,500
Output Tokens Per Call	~5,200
Total Input Tokens / Month	~12,000,000 (12 M)
Total Output Tokens / Month	~8,300,000 (8.3 M)

2. Token Breakdown Per Request

System Prompt (scoring checklists)	~6,500
User Message (topic + essay)	~1,000
Total Input Per Call	~7,500
JSON Response (all analysis sections)	~5,000
Model Essay (requested ~50% of calls)	~400
Average Output Per Call	~5,200

3. Google Gemini Pricing

Gemini 2.5 Flash (CURRENT)	\$0.075	\$0.30	~\$3.50
Gemini 2.5 Pro	\$1.25	\$10.00	~\$98.00

* Gemini 2.5 Flash free tier: 500 requests/day = 15,000/month. SmartLabs current volume (1,600/month) fits entirely within the free tier — potential monthly cost = \$0.

4. OpenAI / ChatGPT Pricing

GPT-4o mini	\$0.15	\$0.60	~\$7.00

o3-mini	\$1.10	\$4.40	~\$50.00
GPT-4o	\$2.50	\$10.00	~\$113.00

5. Anthropic / Claude Pricing

Claude 3.5 Haiku	\$0.80	\$4.00	~\$43.00
Claude 3.7 Sonnet	\$3.00	\$15.00	~\$161.00
Claude 3 Opus	\$15.00	\$75.00	~\$803.00

6. Full Comparison — All Providers

1	Gemini 2.5 Flash	~\$3.50 (possibly \$0)	CURRENT — Best value
2	GPT-4o mini	~\$7.00	Good quality, budget option
3	Claude 3.5 Haiku	~\$43.00	Good quality
4	o3-mini	~\$50.00	Strong reasoning
5	Gemini 2.5 Pro	~\$98.00	Best Gemini quality
6	GPT-4o	~\$113.00	Top OpenAI model
7	Claude 3.7 Sonnet	~\$161.00	Top Claude model
8	Claude 3 Opus	~\$803.00	Overkill for this use case

7. Scaling Projection

		Gemini 2.5 Flash		
40 students x 40 uses	1,600	~\$3.50	~\$113	~\$161
100 students x 40 uses	4,000	~\$8.75	~\$283	~\$403
200 students x 40 uses	8,000	~\$17.50	~\$565	~\$805
400 students x 40 uses	16,000	~\$35.00	~\$1,130	~\$1,610

8. Recommendation

RECOMMENDATION: Continue using Gemini 2.5 Flash

- Cheapest by a wide margin — 30x cheaper than GPT-4o, 46x cheaper than Claude 3.7 Sonnet.
- Free tier covers current volume — 1,600 calls/month is within the 15,000/month free limit.
- Quality matches GPT-4o and Claude Sonnet for structured JSON essay-scoring tasks.
- Highly scalable — even at 400 students the cost remains only ~\$35/month.
- Already integrated and working in the SmartLabs system.

Upgrade path: *If analysis quality ever needs improvement, switch to Gemini 2.5 Pro (~\$98/month) before considering any other provider.*